

Model Analysis Report

Insurance Premium Model

MKM-IP-001

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Governance Framework: SR 11-7 / SS1/23 Model Risk Management

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1 Sensitivity Analysis

This section presents the sensitivity of model outputs to key input parameters. Parameter perturbation ranges are chosen to cover plausible operating conditions and stress scenarios as required under SR 11-7 guidance.

Table 1: Premium sensitivity to flood risk factor (base property £250k).

Flood factor	Premium (£)
0.5000	437.5
0.7500	656.2
1.00	875.0
1.50	1,312.5
2.00	1,750.0
3.00	2,625.0
5.00	4,375.0

Table 2: Premium sensitivity to property value (all factors at 1.0).

Property value (£)	Premium (£)
100,000	350.0
150,000	525.0
200,000	700.0
250,000	875.0
350,000	1,225.0
500,000	1,750.0
750,000	2,625.0

Table 3: Premium sensitivity to claims history factor.

Claims factor	Premium (£)
0.8000	700.0
0.9000	787.5
1.00	875.0
1.20	1,050.0
1.50	1,312.5
2.00	1,750.0
2.50	2,187.5

Table 4: Premium sensitivity to flood factor $\times$ age factor interaction.

Flood factor	Age factor	Premium (£)
1.00	0.8000	700.0
1.00	1.00	875.0
1.00	1.30	1,137.5
1.00	1.60	1,400.0
2.00	0.8000	1,400.0
2.00	1.00	1,750.0
2.00	1.30	2,275.0
2.00	1.60	2,800.0
3.00	0.8000	2,100.0
3.00	1.00	2,625.0
3.00	1.30	3,412.5
3.00	1.60	4,200.0

2 Stress Testing

This section documents stress testing scenarios applied to the model under extreme but plausible conditions.

2.1 Stress Scenarios

Table 5: Stress testing scenarios for Insurance Premium Model.

Scenario	Parameter Shock	Severity	Status
Baseline	No shock	—	Passed
Moderate stress	+2 σ inputs	Medium	Pending
Severe stress	+3 σ inputs	High	Pending
Reverse stress	Output threshold breach	Critical	Pending

Detailed stress test results will be populated as scenarios are executed. Refer to the model's validation plan for the full stress testing programme.

3 Model Performance

This section tracks key performance metrics for the model across validation and production environments.

Table 6: Performance metrics for Insurance Premium Model.

Metric	Value	Status
Unit test pass rate	See Test Results	—
Execution time (single run)	TBD	—
Numerical stability	TBD	—
Reproducibility	Deterministic	OK

This report is produced automatically; human review is required before formal model approval. Supporting artefacts are available in the model documentation directory.